RAI-818: INTELLIGENT MOBILE ROBOTICS

Road Maps

- Voronoi Diagrams

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:
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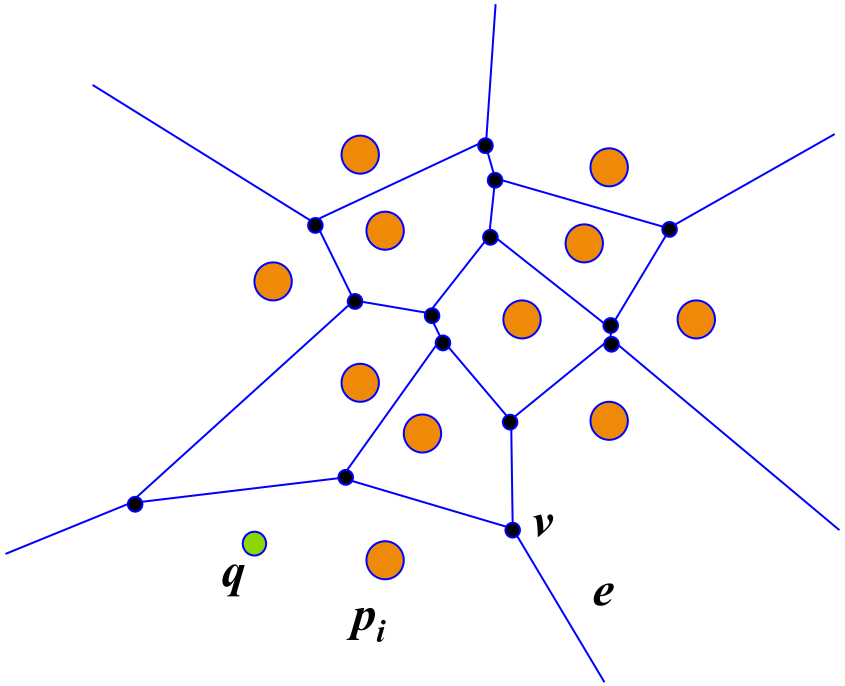
- Roadmaps
  - Voronoi Diagrams
    - Generalized Voronoi Diagrams (GVDs)

Generalized Voronoi Diagram (GVD)

- The ***Generalized Voronoi Diagram (GVD)*** is a set of points where the distance to the two closest obstacles is the same.

- ***Voronoi Diagram*** is defined for a set of points called ***sites***
- Let P be a set of n distinct points (**sites**) in the plane
- ***Voronoi Region*** is the set of points closer to a particular site
- ***Voronoi Diagram*** is the set of points equidistant fromtwo sites
- This will be the subdivision of the plane into n cells, one for each ***site***
- In the planar case ***Voronoi Diagram*** is a collection of line segments



p_i : site points v : Voronoi vertex
q : free point e : Voronoi edge

Defining futures

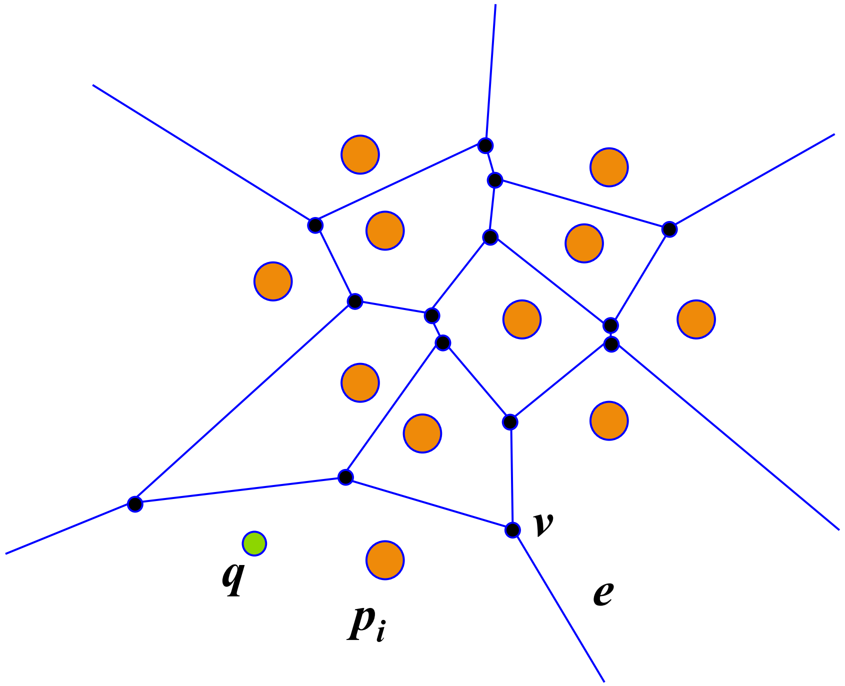
Generalized Voronoi Diagram (GVD)

- **For the purpose of path planning we think of *sites* as *obstacles***
- **Therefore the generalized *Voronoi Region F_i* is the set of points closest to QO_i**

$$F_i = \{q \in Q_{free} \mid d_i(q) \le d_h(q) \quad \forall h \ne i\}$$

where $d_i(q)$ is the distance to an obstacle QO_i from q

$$d_i(q) = \min_{c \in QO_i} d(q,c)$$



p_i : site points **v : Voronoi vertex**
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Defining futures

Generalized Voronoi Diagram (GVD)

- The basic building block for the **GVD** is the set of points equidistant from two obstacles QO_i and QO_j known as a ***two-equidistant surface***

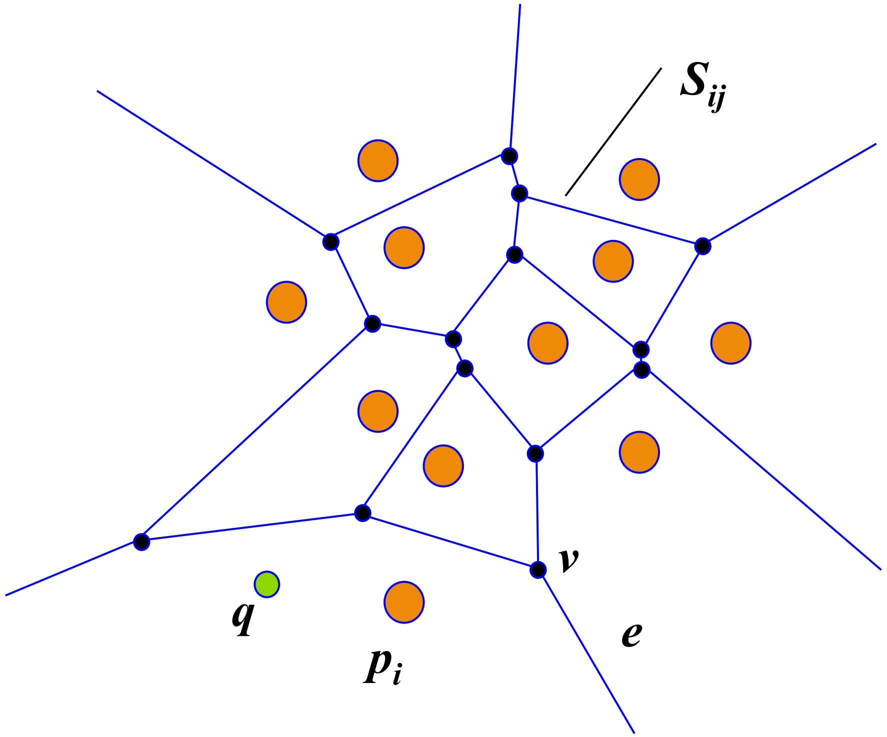
$$S_{ij} = \{x \in Q \mid (d_i(q) - d_j(q)) = 0\}$$

- This surface pierces obstacles so we restrict it to a ***two-equidistant face***

$$F_{ij} = \{q \in S_{ij} \mid d_i(q) \leq d_h(q) \quad \forall h\}$$

- Thus the union of ***two-equidistant faces*** makes the GVD

$$GVD = \bigcup_i \bigcup_j F_{ij}$$



p_i : site points v : Voronoi vertex
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- Roadmaps
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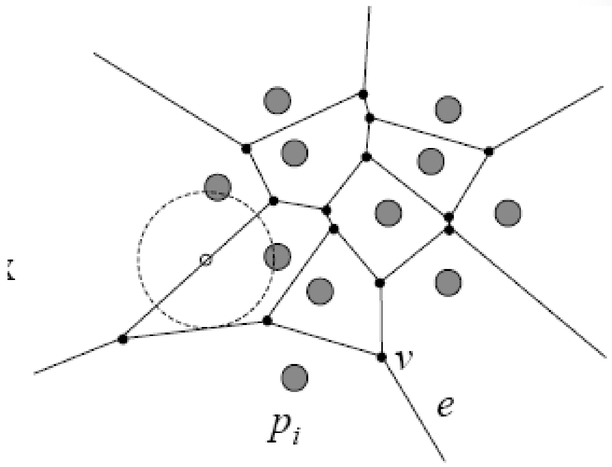
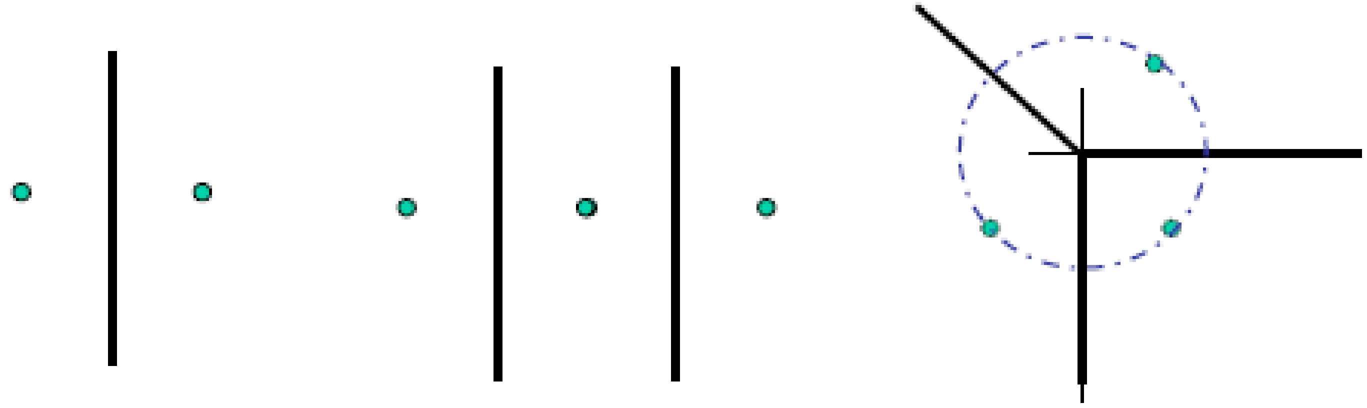
Generalized Voronoi Diagram (GVD)

- A point q lies on a *Voronoi Edge* between sites p_i and p_j *iff* the largest empty circle centered at q touches only p_i and p_j

- A *Voronoi Edge* is a sub-set of locus of points equidistant from p_i and p_j .



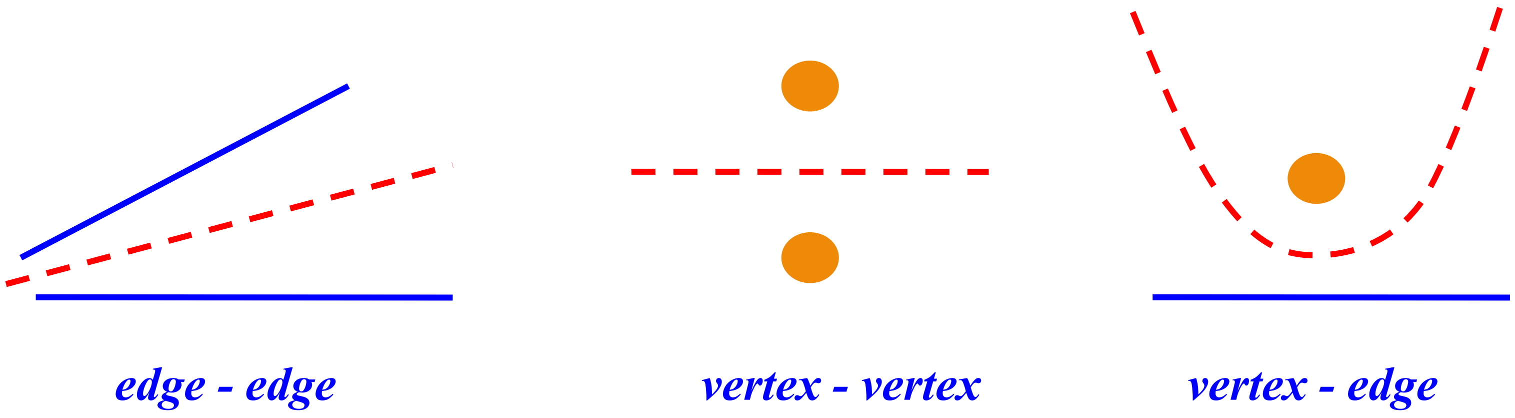


Roadmaps

- **Roadmaps**
  - **Voronoi Diagrams**
    - **Generalized Voronoi Diagrams (GVDs)**

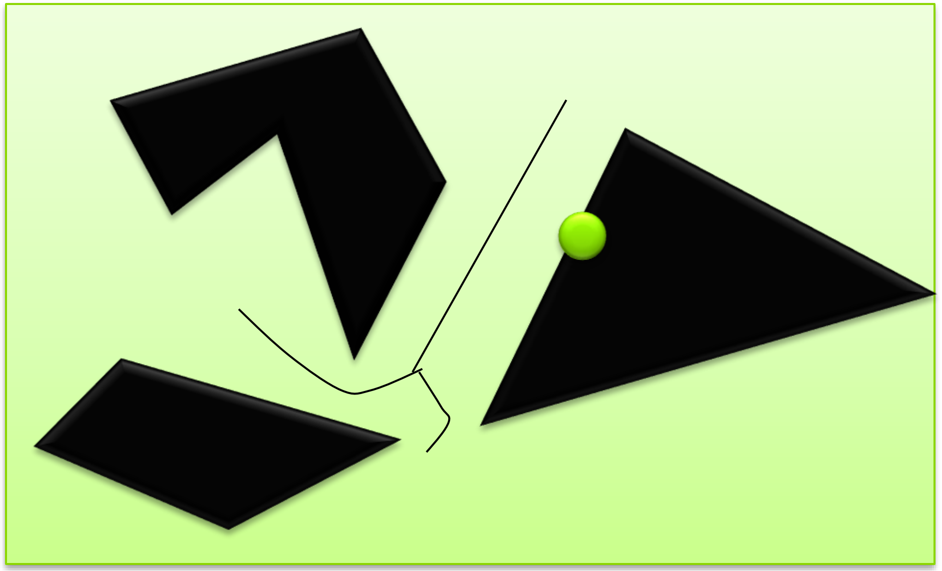
Generalized Voronoi Diagram (GVD)

- **Extending to polygonal objects:**
  - **Features of an obstacle are vertices and edges**
  - **A roadmap edge must be between some pair of features.**



GVD Implementation Bug algorithm

- Online Diagram Generation:
  - Locate two nearest obstacles and move to midpoint
  - Continue along GVG edge until "meet" point
- Detect new GVG edges
- Select one to explore and continue
  - Detect boundary points and cycles



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- Roadmaps
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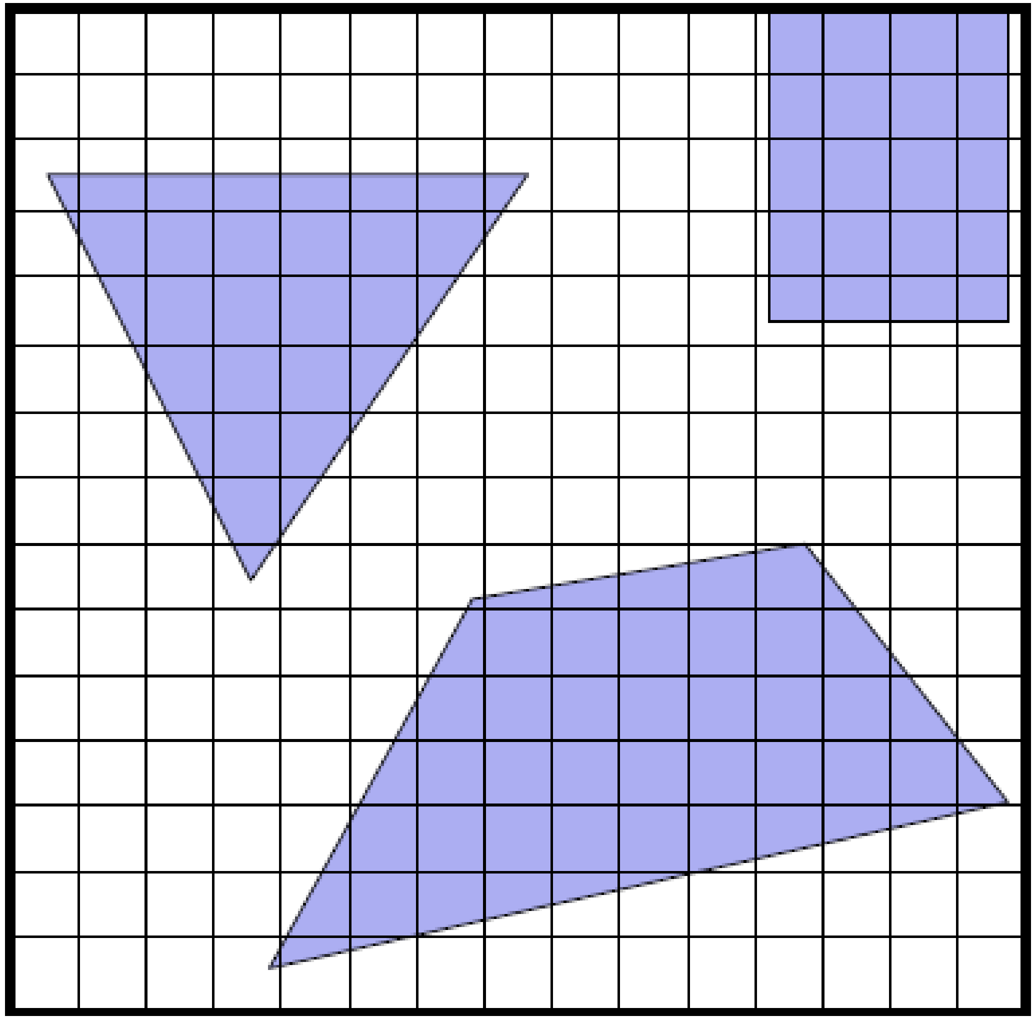
    - GVD Implementation
      - Bug Algorithm
      - Brush Fire Algorithm

GVD Implementation: Brush Fire Algorithm



Defining futures

Potential Fields

- **Roadmaps**
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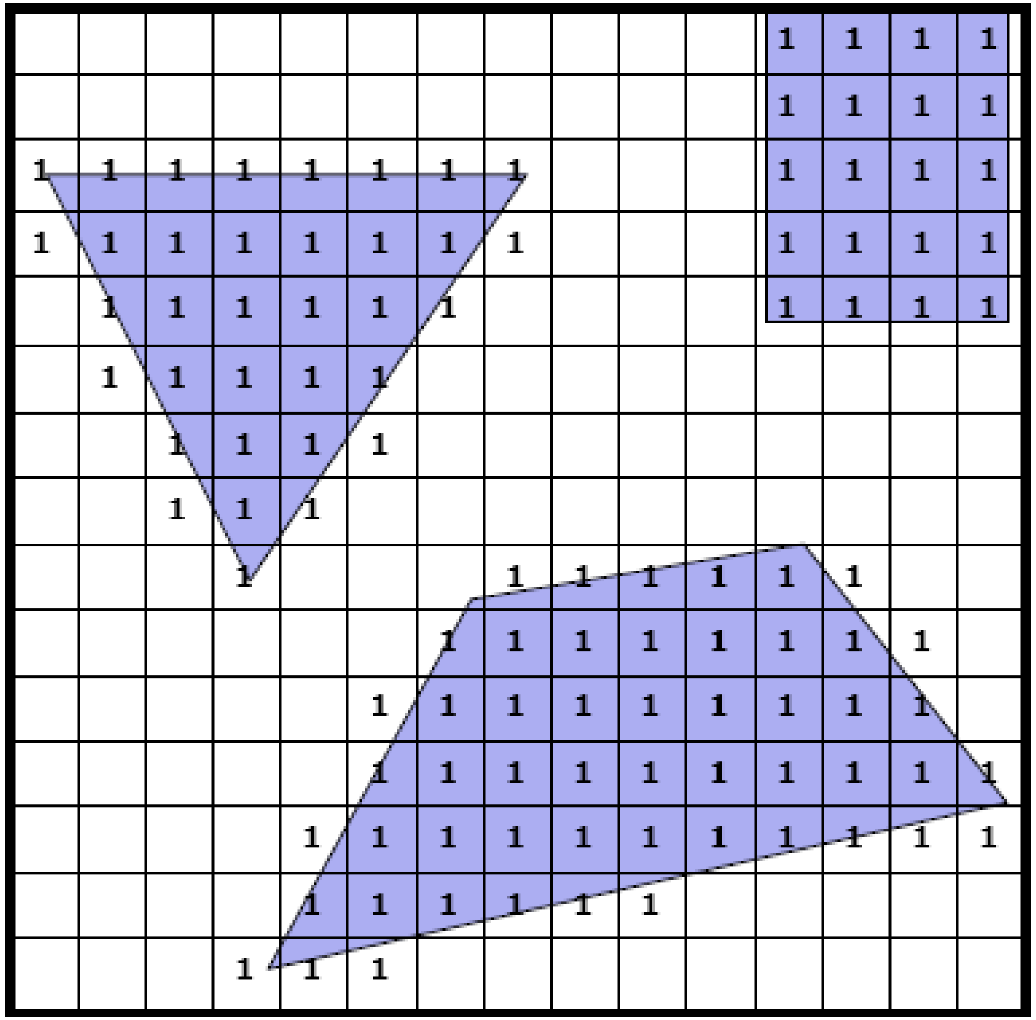
GVD Implementation: Brush Fire Algorithm

- **Start with an empty grid with obstacles equal to 1**



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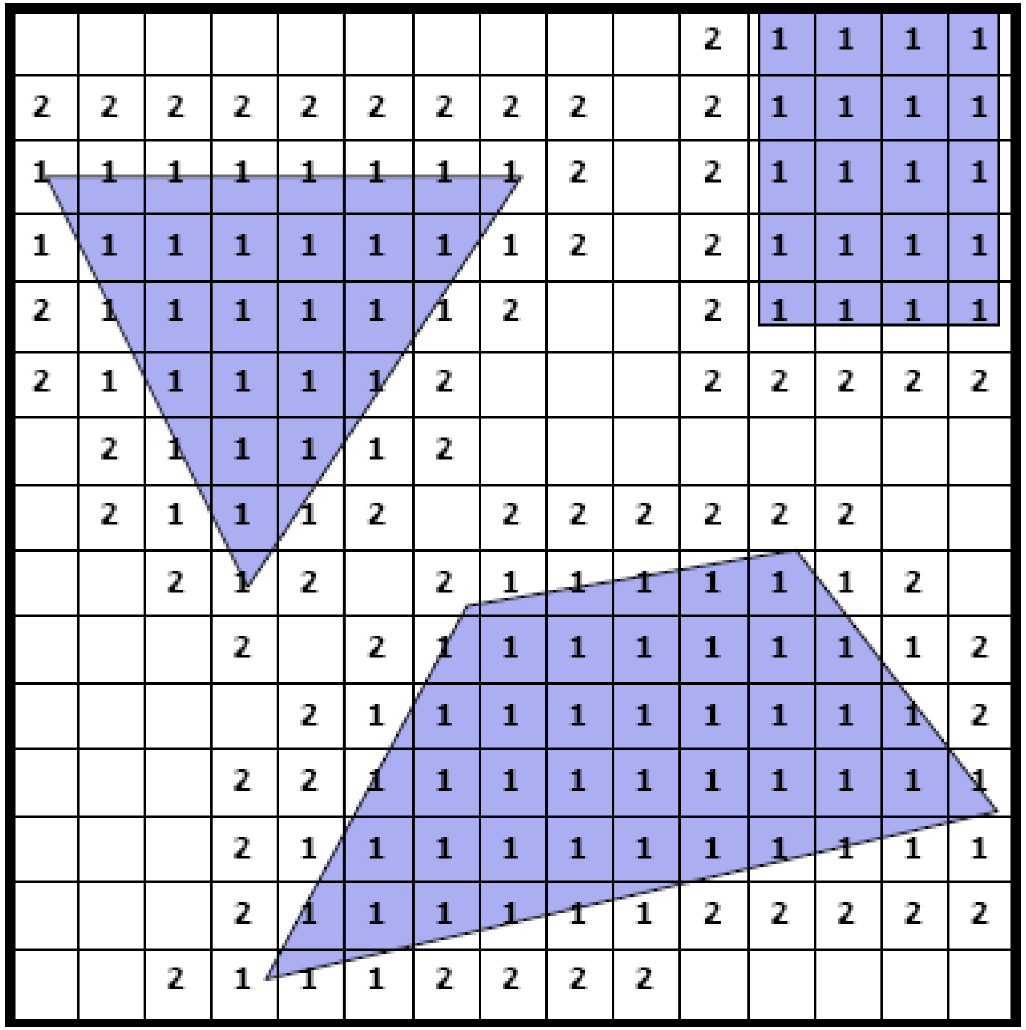
GVD Implementation: Brush Fire Algorithm

- **Start with an empty grid with obstacles equal to 1**
- **Expand cells *i* to *i*+1**



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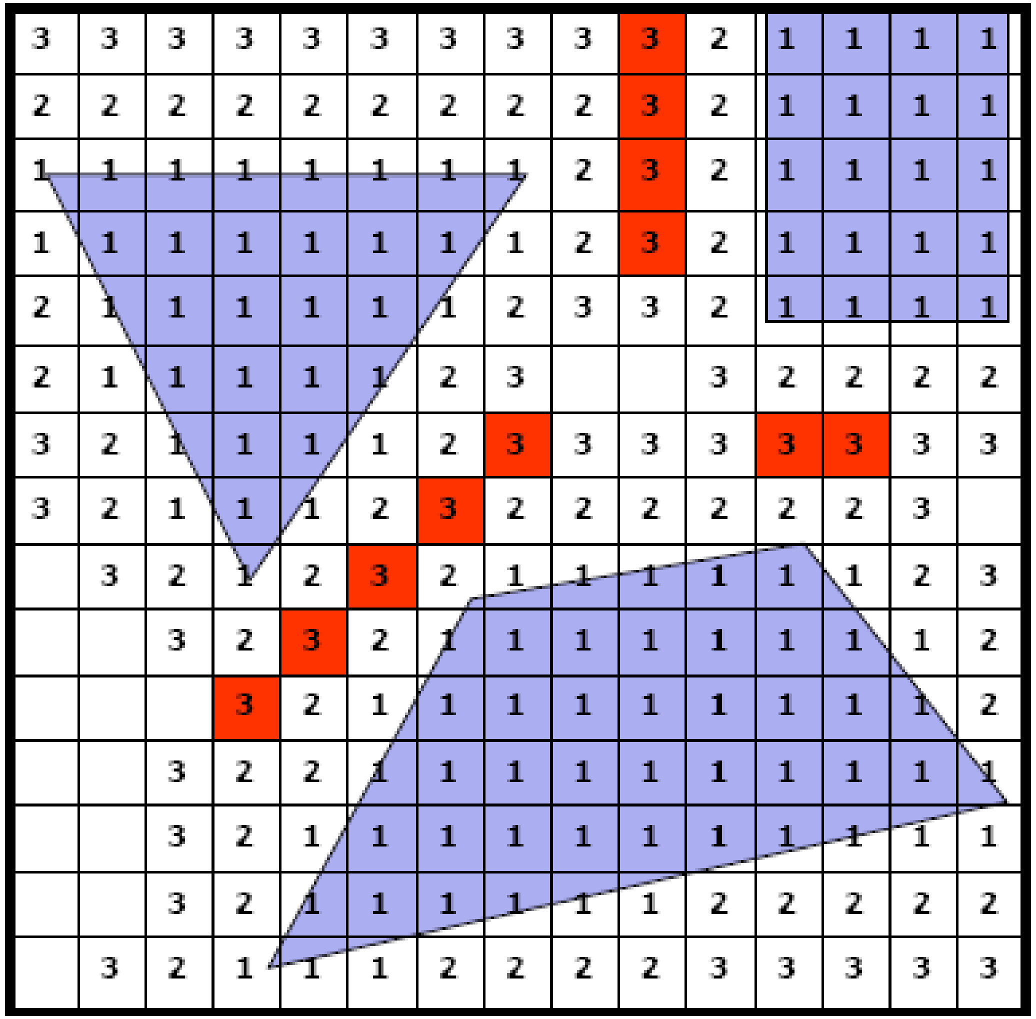
GVD Implementation: Brush Fire Algorithm

- **Start with an empty grid with obstacles equal to 1**
- **Expand cells *i* to *i*+1**
- **Repeat**



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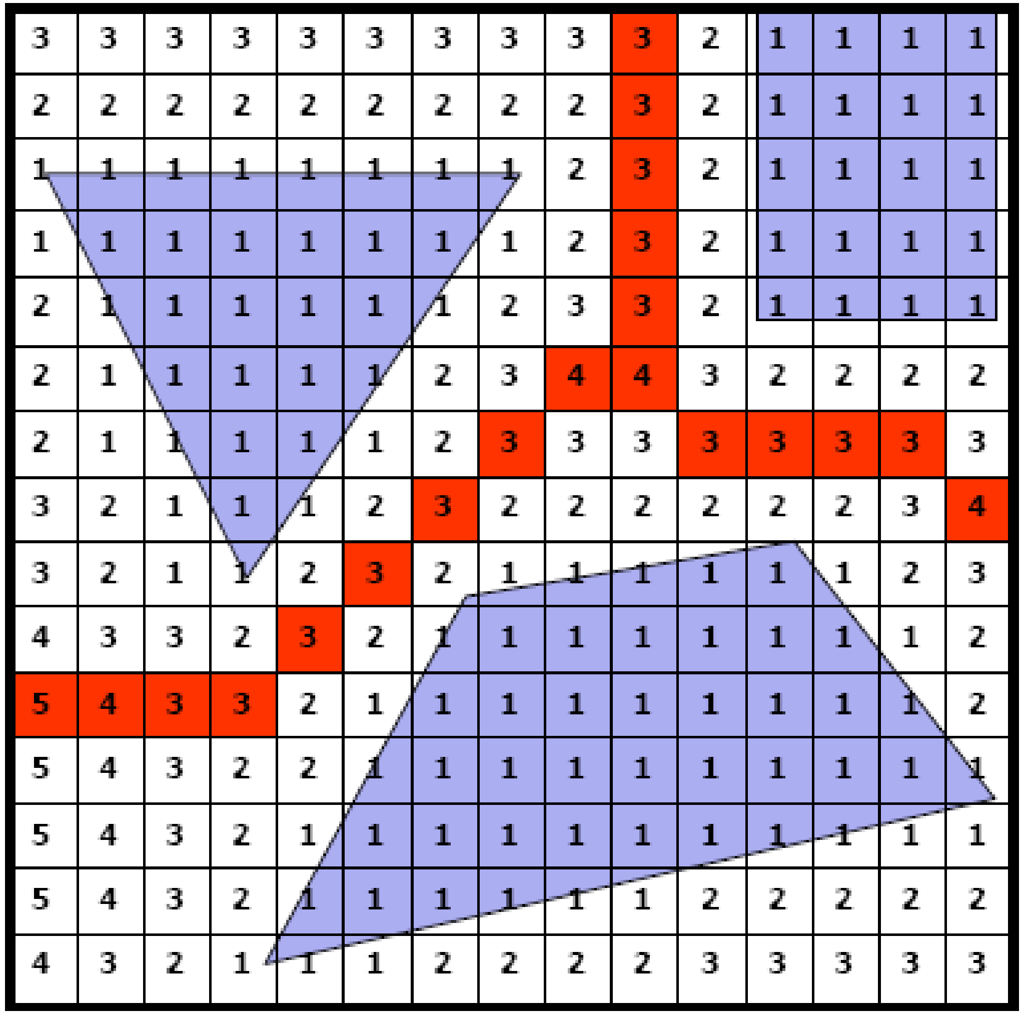
GVD Implementation: Brush Fire Algorithm

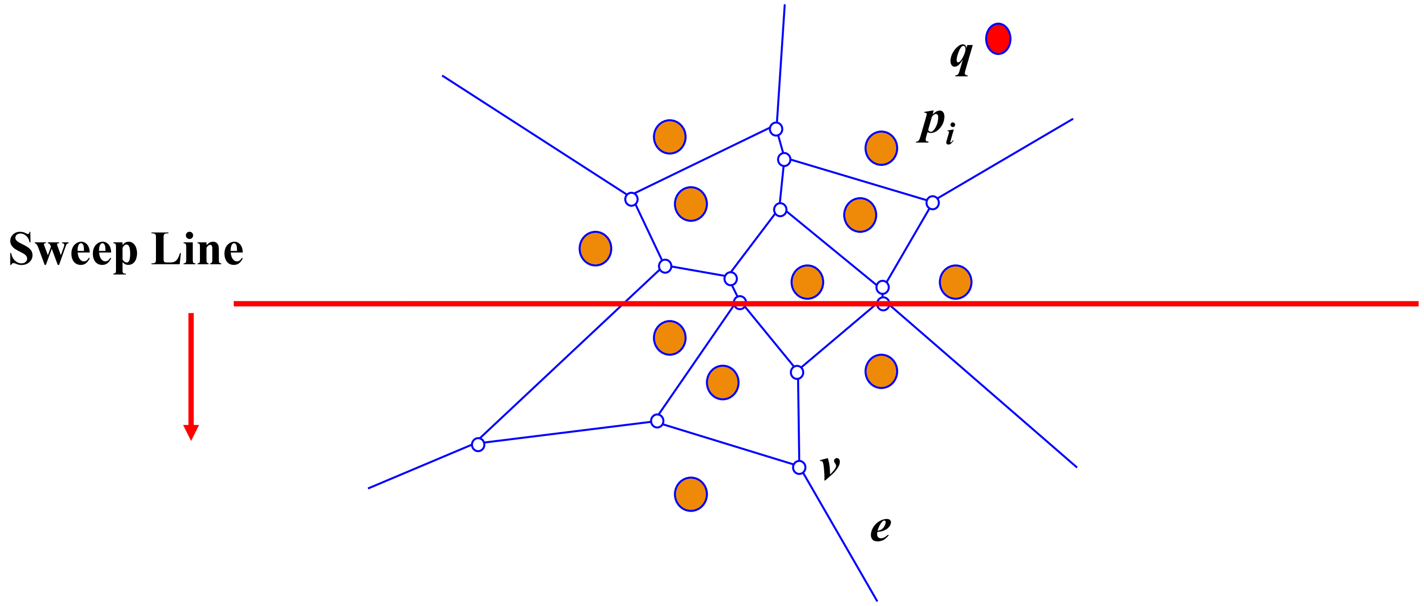
- **Start with an empty grid with obstacles equal to 1**
- **Expand cells *i* to *i*+1**
- **Repeat**
- **Repeat Again**
- **If a cell is expanded twice, label as a GVG edge and do not expand further.**



Defining futures

GVD Implementation: Sweep Line Method

What is the invariant we are looking for?



Maintain a representation of the locus of points q that are closer to some site p_i *above* the sweep line than to the line itself (and thus to any site below the line).

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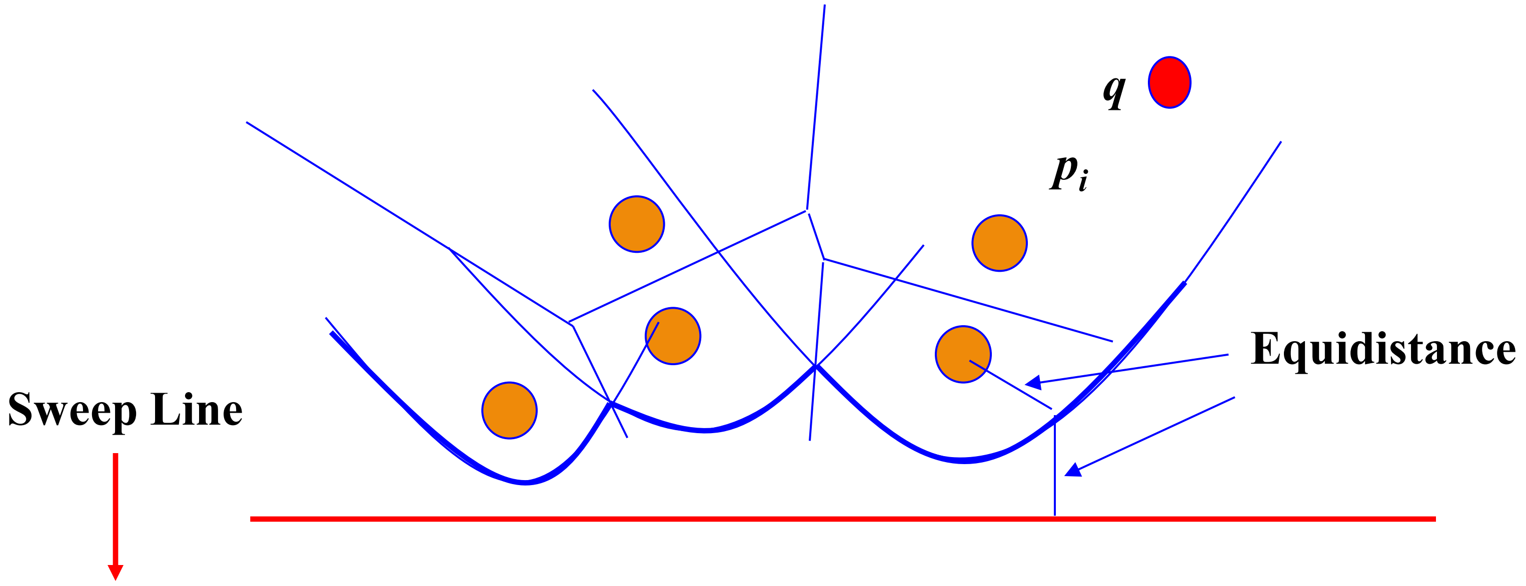
GVD Implementation: Sweep Line Method

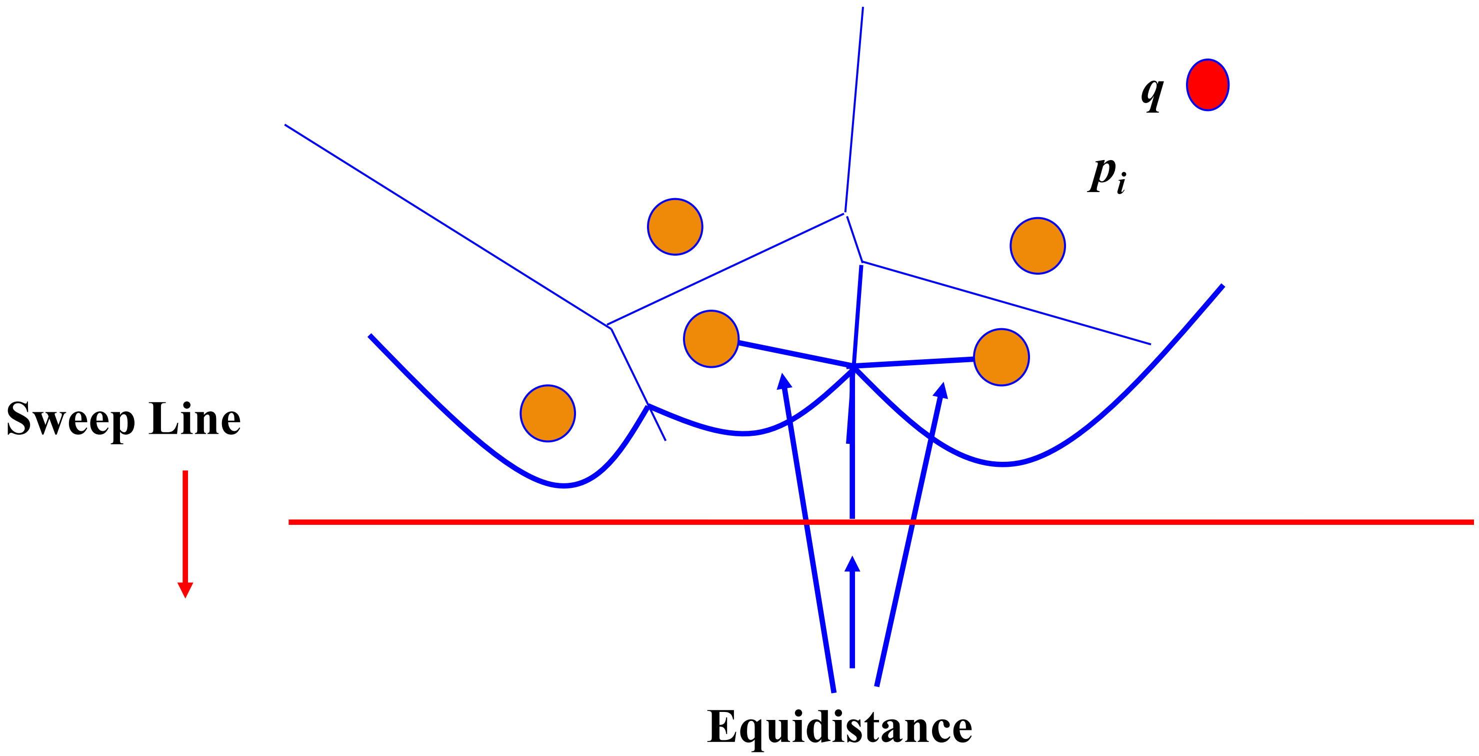
Which points are closer to a site above the sweep line than to the sweep line itself?



The set of parabolic arcs form a beach-line that bounds the locus of all such points

GVD Implementation: Sweep Line Method

Break points trace out Voronoi Edges.



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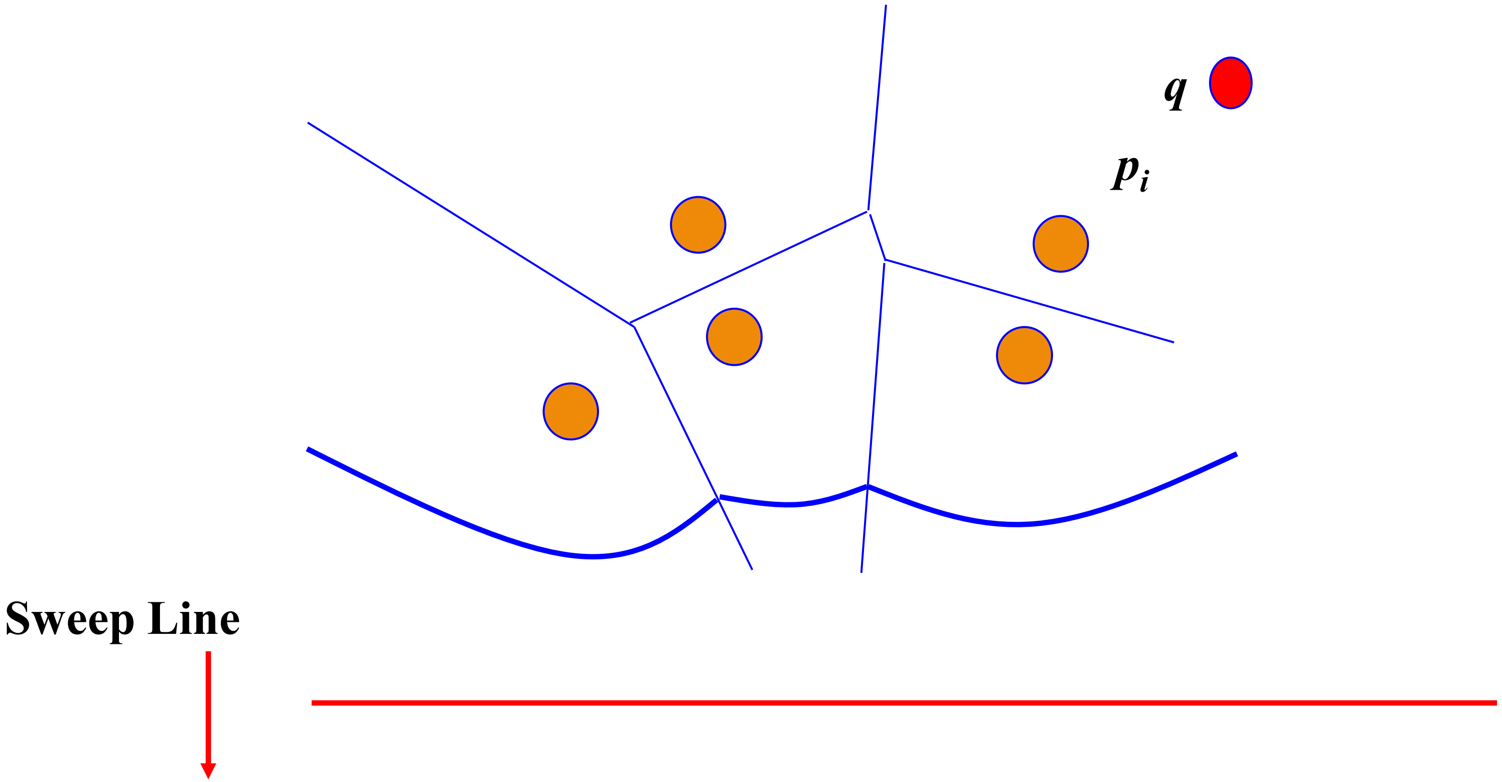
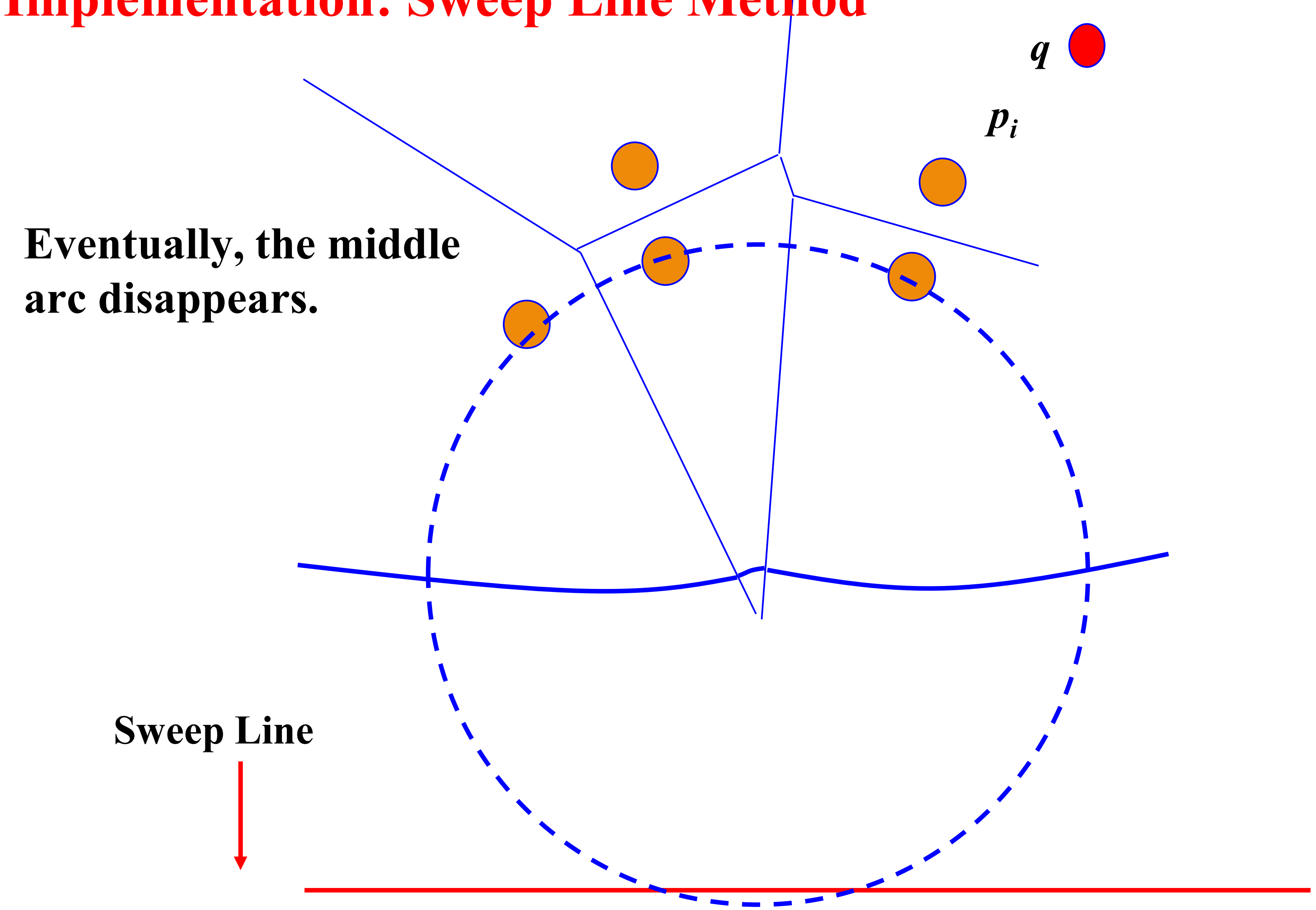
GVD Implementation: Sweep Line Method



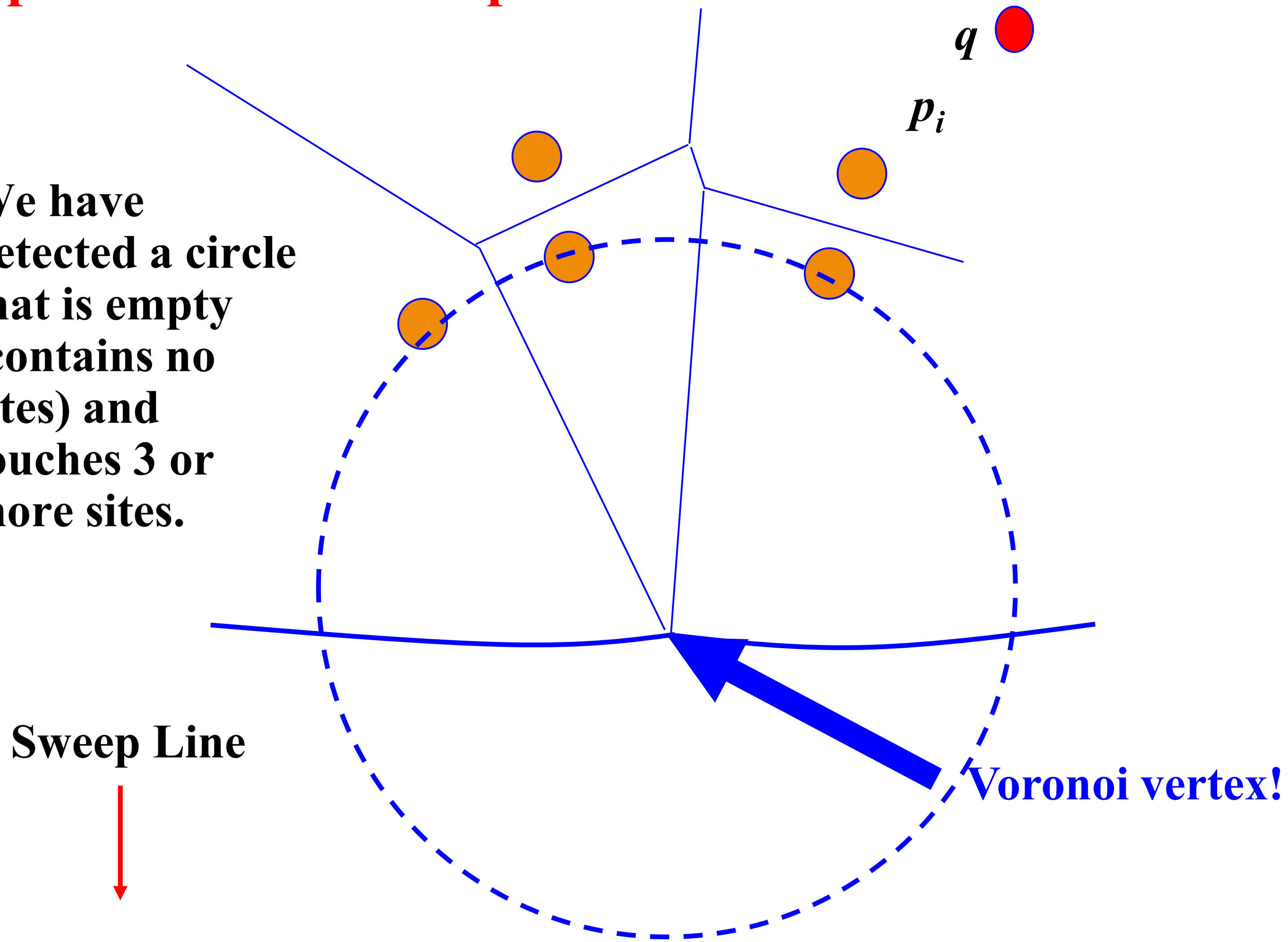
Arcs flatten out as sweep line moves down.

GVD Implementation: Sweep Line Method



GVD Implementation: Sweep Line Method

We have detected a circle that is empty (contains no sites) and touches 3 or more sites.



Defining futures

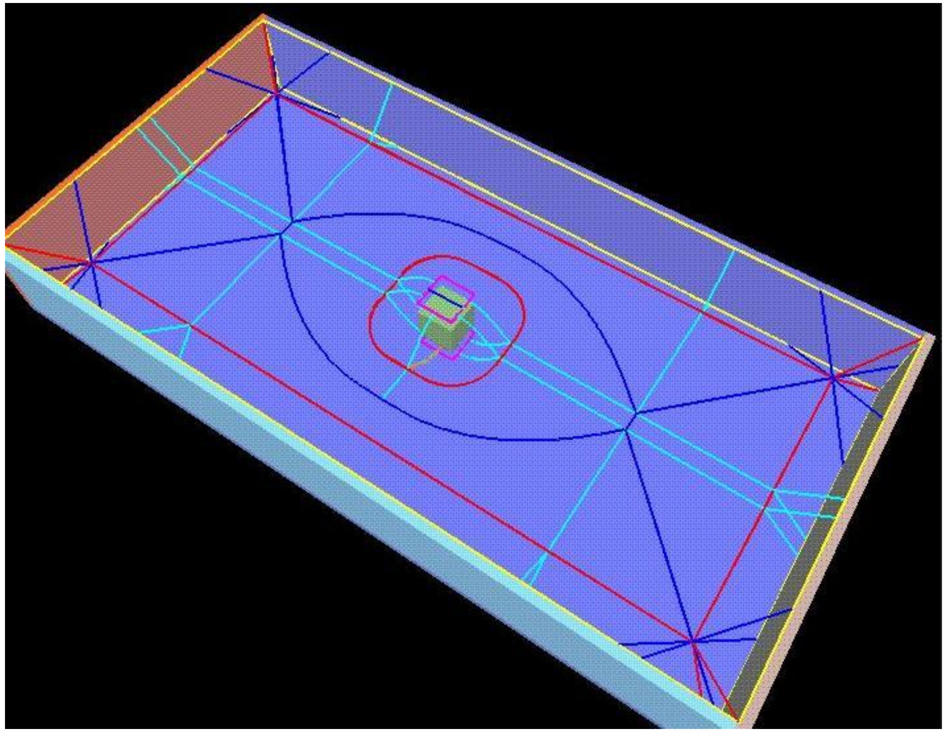
GVD Implementation: Beach Line Properties

- *Voronoi Edges* are traced by the break points as the *sweep line* moves down.
  - Emergence of a new *break point(s)* (from formation of a new arc or a fusion of two existing *break points*) identifies a new *edge*
- *Voronoi Vertices* are identified when two break points meet (fuse).
  - Decimation of an old arc identifies new vertex

Defining futures

HGVG as Roadmap

HGVG is a roadmap, constructed incrementally using sensor data.



HGVG as Roadmap : Assumptions

- Robot is a point
- Workspace contains only convex obstacles
- Non-convex obstacles are modeled as the union of convex obstacles
- Bounded space

Defining futures

- **Roadmaps**
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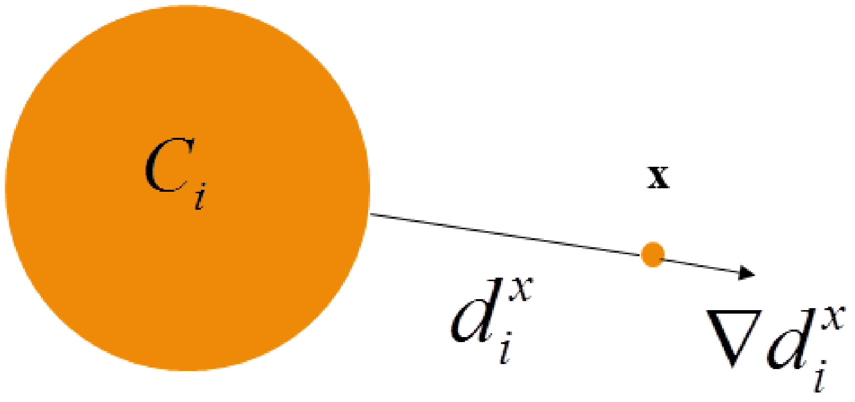
    - GVD Implementation
    - Hierarchical Generalized Voronoi Graphs (HGVGs)
      - Assumptions
      - Basic Definitions

HGVG as Roadmap : Basic Definitions

Single-Object Distance Function

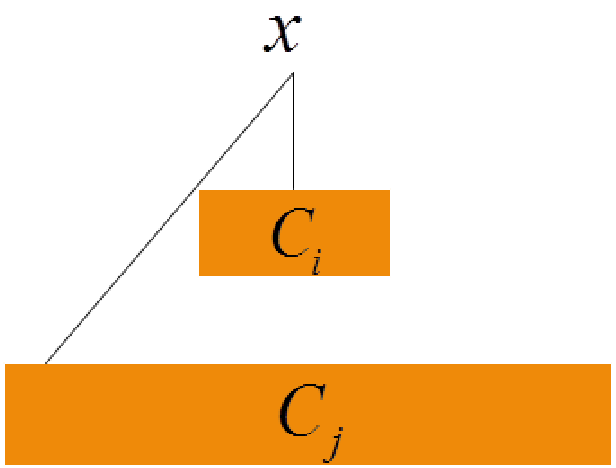


$$d_i^x(x) = \min_{c_0 \in C_i} \| x - c_0 \|$$

$$\nabla d_i^x(x) = \frac{x - c_0}{\| x - c_0 \|}$$

Multi-Object Distance Function

$$D^x(x) = \min_i d_i^x(x)$$



Defining futures

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      - Basic Definitions
      - Two Equidistant Surfaces

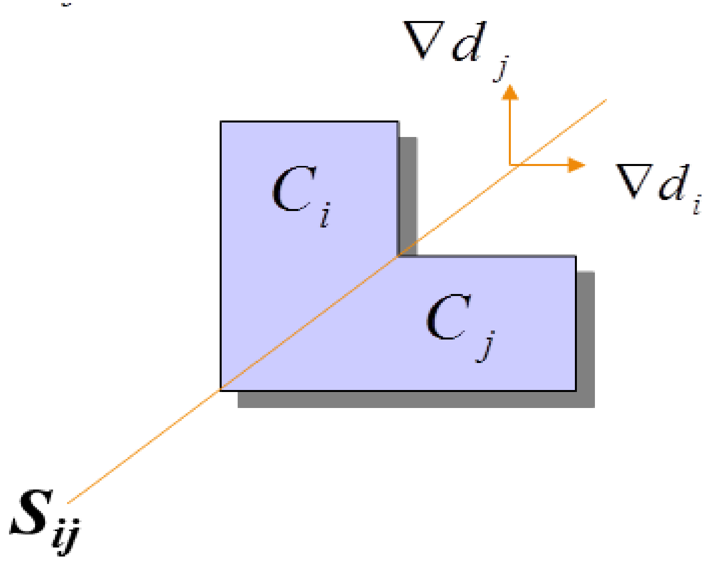
Two-Equidistant Surfaces

- **Two Equidistant Surface**

$$S_{ij} = \left\{x \in Q \mid \left(d_i(q) - d_j(q)\right) = 0\right\}$$

- **Two Equidistant Surjective Surface**

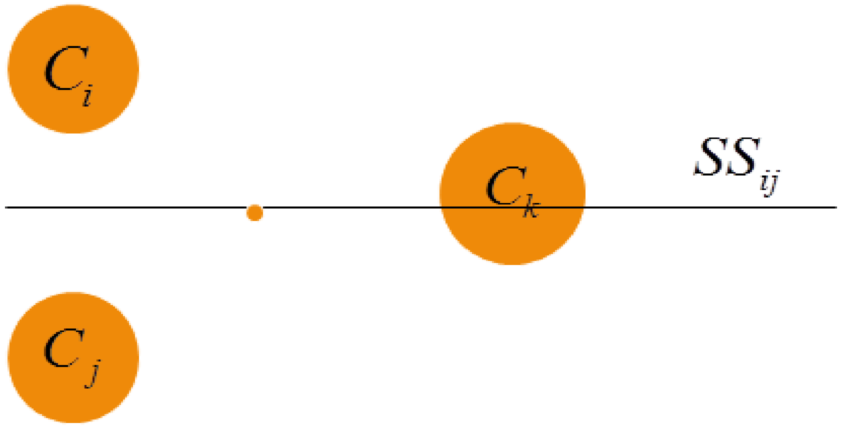
$$SS_{ij} = \left\{x \in S_{ij} \mid \nabla d_i(q) \neq d_j(q)\right\}$$



Defining futures

More Rigorous Definition

- Going through obstacles



- Two Equidistant Face

$$F_{ij} = \{x \in S_{ij} | \nabla d_i(q) = d_j(q) \le d_h(q), \forall h \ne i, j\}$$

Defining futures

Roadmaps

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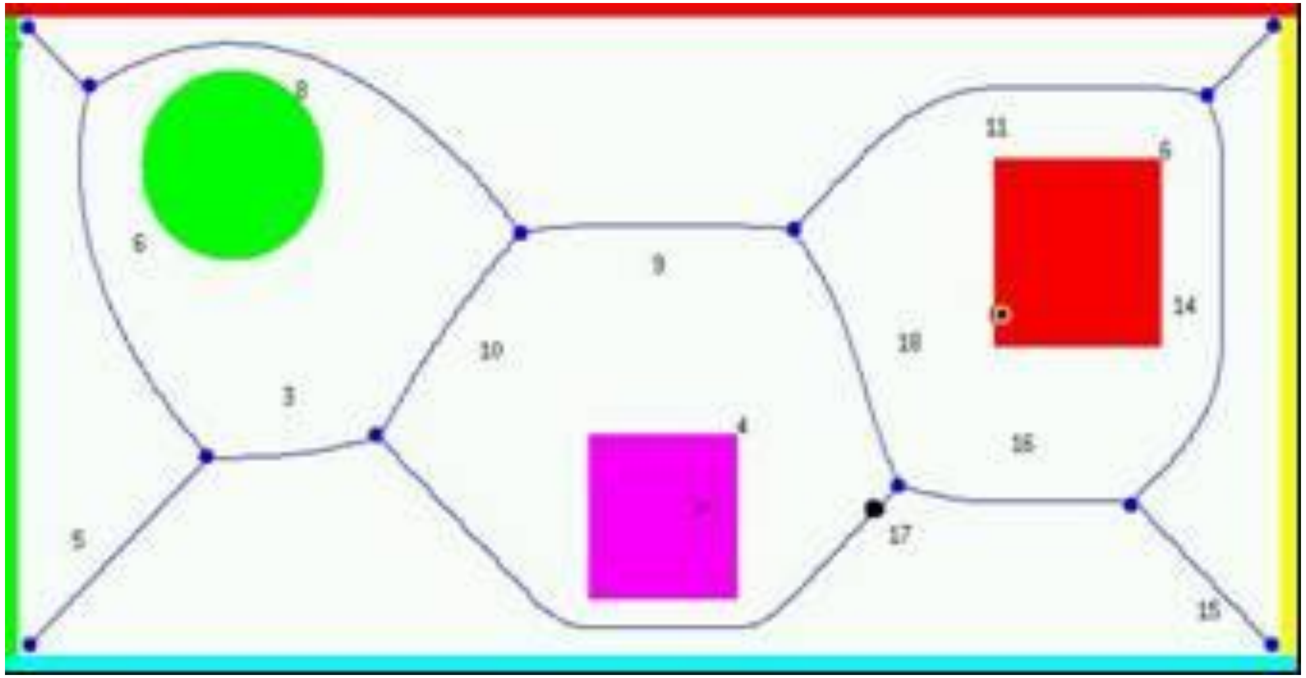
      - Assumptions
      - Basic Definitions
      - Two Equidistant Surfaces
      - **Examples**

Defining futures

General Voronoi Diagram

$$\mathrm{GVD} = \bigcup_{i=1}^{n-1} \bigcup_{j=i+1}^{n} F_{ij}$$



Defining futures

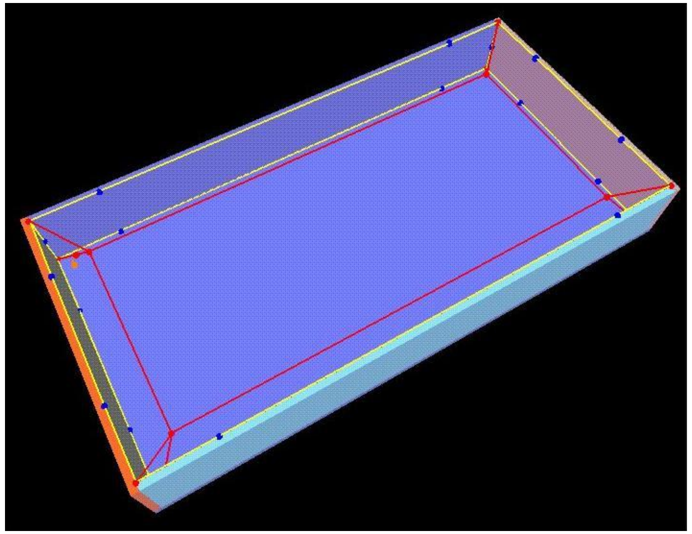
General Voronoi Graph

- **In 3-Dimensions**

$$F_{ijk} = F_{ij} \cap F_{ik} \cap F_{jk}$$

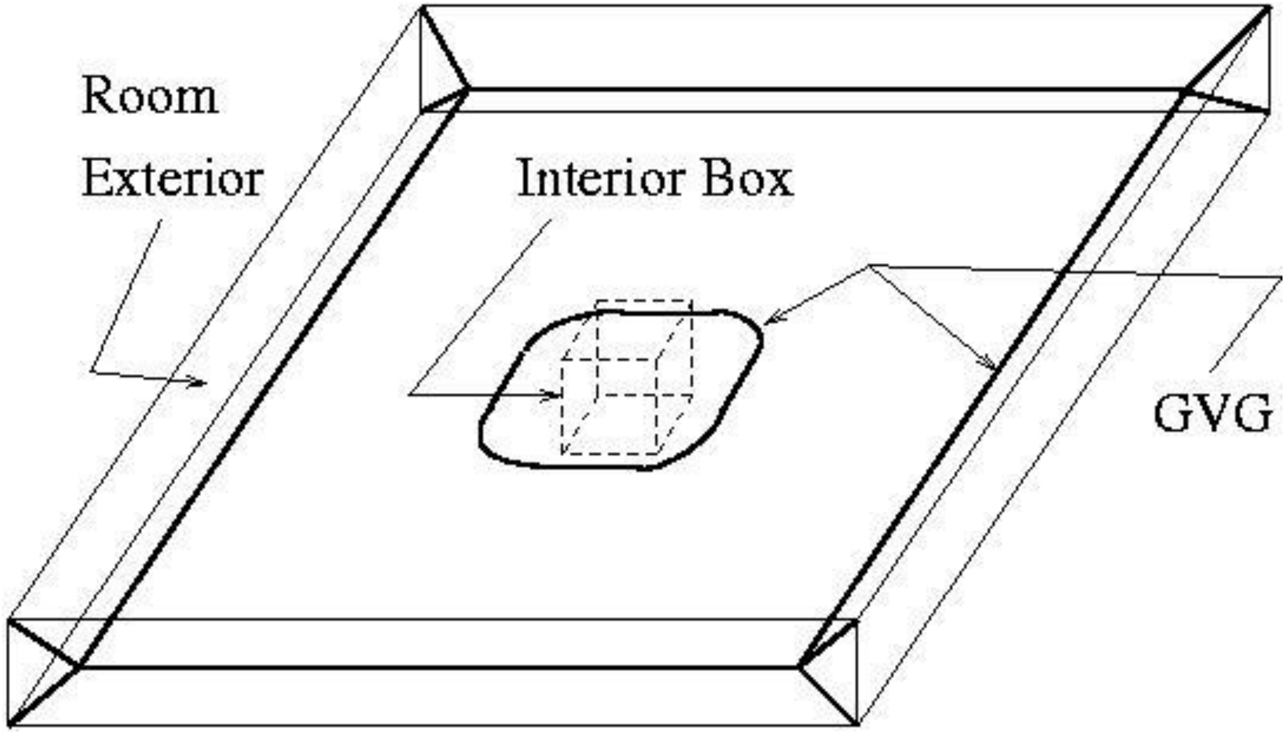
- **In *m*-Dimensions**

$$F_{ijk\ldots m} = F_{ij} \cap F_{ik} \ldots \cap F_{im}$$
$$= F_{ij\ldots m-1} \ldots \cap F_{im}$$



GVG Connected?

∂F_{ij} **is not connected**



Defining futures

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GVG²

- ***Second-order two-equidistant surface***

$$F_k |_{F_{ij}} = \{x \in F_{ij} : \forall h \neq i, j, k,$$
$$d_h(x) > d_k(x) > d_i(x) = d_j(x) > 0$$
$$\text{and } \nabla d_i \neq \nabla d_j\}$$